

GeoRisk-SPC: Geometry-conditioned Pseudo-label Risk Supervision for Semi-supervised Surgical Instrument Segmentation

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Abstract—Surgical instrument segmentation is a key prerequisite for surgical scene understanding, robotic perception, and image-guided assistance, yet dense annotation remains expensive because tool boundaries are thin, reflective, articulated, and frequently occluded. Semi-supervised learning reduces this burden by exploiting unlabeled frames, but its effectiveness is limited by a structured failure mode: pseudo-label reliability is spatially heterogeneous, with trustworthy predictions in homogeneous tool interiors coexisting with unreliable predictions around boundaries, specular highlights, occlusions, and depth discontinuities. Existing semi-supervised segmentation methods mainly improve pseudo-label generation or consistency regularization, but they do not explicitly decide which surgical pixels should be trusted by hard pseudo-label supervision. We propose GeoRisk-SPC, a geometry-conditioned pseudo-label risk supervision framework for semi-supervised surgical instrument segmentation. GeoRisk-SPC estimates a pixel-wise risk map from monocular depth discontinuity, teacher uncertainty, and geometry-semantic conflict; assigns hard pseudo-label supervision only to low-risk high-confidence pixels; and applies soft structural consistency under risk-guided feature perturbation to high-risk pixels. The full model, GeoRisk-SPC-DG, further uses a depth-guided geometric encoder that injects multi-scale depth and depth-gradient context through gated residual modulation. Experiments on EndoVis 2017 with 4-fold cross-validation show that GeoRisk-SPC-DG achieves 89.78% Dice at 40% labels, 88.09% at 10% labels, and 86.89% at 5% labels, outperforming strong recent baselines including UniMatch and SegMatch under the same protocol. GeoRisk-SPC-DG with 10% labels also surpasses SegMatch with 40% labels, indicating that geometry-aware risk supervision improves label efficiency by suppressing error propagation from unreliable pseudo-label regions.

Index Terms—semi-supervised medical image segmentation, surgical instrument segmentation, pseudo-label reliability, geometric risk localization, depth-guided segmentation

I. INTRODUCTION

Robotic-assisted minimally invasive surgery is increasingly data-driven. Accurate surgical instrument segmentation supports tool tracking, workflow analysis, human-robot interaction, augmented reality guidance, and safety-critical scene understanding [1], [2], [3]. Modern segmentation networks have made substantial progress, yet their deployment in surgical video remains constrained by annotation cost. Pixel-wise instrument masks require expert review, careful boundary tracing, and repeated quality control because tools are thin, articulated, reflective, and frequently occluded by tissue, smoke, blood, or other instruments. Recent work has

therefore explored synthetic data, active learning, and semi-supervised learning to reduce the annotation burden [4], [5], [6], [7]. Semi-supervised learning (SSL) is appealing for surgical video because unlabeled frames are abundant. Most SSL segmentation methods combine pseudo-labeling and consistency regularization: a teacher or a peer model predicts labels on weakly augmented images, and a student is trained to match those predictions under stronger perturbations [8], [9], [10], [11], [12]. Medical image segmentation has followed the same trajectory, progressing from uncertainty-aware teacher-student learning [13], [14] to stronger TMI-era paradigms based on cross-style consistency, adversarial reinforcement learning, probabilistic prototypes, and versatile collaborative training [15], [16], [17], [18]. These methods improve how pseudo-labels are generated, filtered, propagated, or regularized. However, surgical instrument segmentation exposes a limitation that remains under-addressed in this line of work: pseudo-label errors are not spatially uniform. The interior of a metallic instrument shaft can be predicted with high confidence, while its boundary, tip, reflection, or occlusion interface remains unreliable in the same frame. Treating all confident unlabeled pixels with a single rule ignores this heterogeneity. A globally high confidence score can still be misleading near fine boundaries, and uniformly applying hard pseudo-label supervision can reinforce boundary leakage, tool fragmentation, and foreground-background confusion. This failure mode is especially relevant for surgical scenes, where visual ambiguity is coupled with geometric discontinuities caused by tool-tissue separation, articulated tool motion, and abrupt depth changes. The key observation of this work is that unreliable pseudo-label regions are geometrically structured. Monocular depth estimation has matured into a practical dense prediction tool [19], and depth-aware segmentation studies show that depth cues are useful for disambiguating object boundaries and scene layout [20], [21], [22]. In surgical video, depth discontinuities provide an external cue that is complementary to teacher confidence: an uncertain teacher prediction located on a depth edge is more suspicious than the same uncertainty in a homogeneous region, while disagreement between depth boundaries and predicted semantic boundaries reveals geometry-semantic conflict. This motivates a shift from making pseudo-labels uniformly stronger to estimating where pseudo-labels should be trusted. We propose GeoRisk-SPC (Geometry-conditioned Pseudo-label Risk Supervision), a semi-supervised framework that explicitly models pixel-wise pseudo-label risk for surgical instrument segmentation.

GeoRisk-SPC first estimates a risk map by combining local depth discontinuity, teacher entropy, and geometry-semantic conflict. It then uses the risk map to partition unlabeled pixels into low-risk high-confidence regions and high-risk ambiguous regions. Low-risk pixels receive hard pseudo-label supervision, whereas high-risk pixels are trained through soft structural consistency between clean and perturbed student branches. A risk-guided feature perturbation module concentrates representation disturbance on high-risk regions so that robustness is learned where pseudo-label errors occur. The full variant, GeoRisk-SPC-DG, uses a depth-guided geometric encoder to inject multi-scale depth and depth-gradient context into RGB features through gated residual modulation. Our key contributions are:

- We formulate semi-supervised surgical instrument segmentation as a geometry-conditioned pseudo-label risk problem, replacing uniform unlabeled-pixel treatment with pixel-wise trust estimation.
- We propose a risk localization module that combines monocular depth discontinuity, teacher uncertainty, and geometry-semantic conflict to identify unreliable pseudo-label regions.
- We design region-aware supervision that assigns hard pseudo-label learning to low-risk pixels and soft structural consistency to high-risk pixels, reducing error propagation while retaining unlabeled learning signals.
- We introduce risk-guided structural perturbation and a depth-guided geometric encoder that injects multi-scale depth context into RGB features through gated residual modulation.
- We conduct 4-fold evaluation on EndoVis 2017 under 5%, 10%, 20%, and 40% label ratios, comparing against classical SSL methods, recent semantic segmentation baselines, surgical SSL methods, and available official implementations.

II. RELATED WORK

A. Surgical Instrument Segmentation

Surgical instrument segmentation has been studied as a core component of surgical data science and computer-assisted intervention [1], [3]. EndoVis challenge datasets have accelerated benchmarking, while recent methods address domain shift, limited labels, and realistic image generation [2], [23], [4], [5]. Compared with natural-scene segmentation, surgical instrument segmentation has distinctive failure cases: instruments are thin, reflective, deformable in appearance due to articulation, and visually entangled with tissue. These properties make dense annotation expensive and make boundary errors clinically relevant. Our work targets this label-efficiency problem with a semi-supervised framework designed around boundary and geometry-dependent pseudo-label risk.

B. Semi-supervised Semantic Segmentation

Semi-supervised semantic segmentation reduces annotation requirements by leveraging unlabeled images. Mean Teacher [8] stabilizes consistency targets using an EMA teacher. CPS

[10] introduces cross pseudo supervision between two networks. U2PL [11] explicitly uses unreliable pseudo-labels as negative samples for contrastive learning. UniMatch and UniMatch V2 revisit weak-to-strong consistency with unified pseudo-label training [12], [24], and CorrMatch [25] propagates labels through correlation matching. These methods are strong generic baselines, but their reliability mechanisms are driven mainly by confidence, augmentation, or feature correlation. They do not use surgical geometry to decide which spatial regions should avoid hard pseudo-label supervision.

C. Semi-supervised Medical and Surgical Segmentation

Medical SSL segmentation has developed a rich set of uncertainty, consistency, and prototype-based methods. UAMT [13] uses uncertainty-aware self-ensembling, and URPC [14] applies uncertainty-rectified pyramid consistency. More recent TMI and MICCAI works strengthen SSL through cross-style consistency with shape and context constraints [15], generative adversarial reinforcement learning [16], probabilistic representation and prototype learning [17], versatile collaborative segmentation [18], and worst-case-aware decoupled training [26]. For surgical instruments, MMS [6] introduces a contrastive semi-supervised network for surgical tool segmentation, Surgivisor [27] uses transformer-based uncertainty-guided supervision, and SegMatch [7] strengthens pseudo-label learning with adversarial augmentation. These studies confirm the importance of unlabeled data but still focus on stronger global consistency or pseudo-label learning. GeoRisk-SPC instead localizes pseudo-label risk at the pixel level and couples the supervision type to geometric reliability.

D. Depth-aware Segmentation

Depth provides geometric cues that are complementary to RGB appearance. Depth-adaptive networks and depth-aware attention show that depth can improve semantic segmentation by emphasizing geometric boundaries and layout [20], [21]. Dense monocular prediction methods such as DPT make depth cues available even when calibrated depth sensors are absent [19]. RGB-D representation learning has continued to advance through stronger fusion designs [22]. In surgical imaging, depth is valuable because tool-tissue interfaces often correspond to geometric discontinuities. Unlike conventional RGB-D segmentation methods that use depth only as an extra feature source, GeoRisk-SPC uses depth discontinuity as a supervisory risk prior for unlabeled pixels.

E. Pseudo-label Quality

Pseudo-label quality is central to SSL. Classic pseudo-labeling [28], MixMatch [9], and FlexMatch [29] improve label assignment through confidence, augmentation, and curriculum learning. Segmentation methods further refine pseudo-labels through uncertainty estimation, negative sampling, correlation propagation, or adversarial perturbation [13], [11], [25], [7]. The remaining gap is spatial trust. A pseudo-label can be reliable in one region and harmful in another within the same image. GeoRisk-SPC addresses this by estimating a pixel-wise risk map and selecting the supervision form according to risk.

Algorithm 1 GeoRisk-SPC Training (Geometry-conditioned Pseudo-label Risk Supervision)

Require: Labeled data $\{(x_l, y_l)\}$, unlabeled data $\{(x_u, d_u)\}$, teacher model T_θ , student model S_ϕ

- 1: **while** not converged **do**
- 2: Sample labeled batch $\{(x_l, y_l)\}$ and unlabeled batch $\{(x_u, d_u)\}$
- 3: Compute teacher prediction $p_t = T_\theta(\text{weak_aug}(x_u))$
- 4: Compute risk map R from d_u and p_t
- 5: Generate masks M_r, M_l from R and teacher confidence
- 6: Student clean: $p_{\text{clean}} = S_\phi(\text{strong_aug}(x_u))$
- 7: Student perturbed: $p_{\text{pert}} = S_\phi(\text{strong_aug}(x_u), M_r)$
- 8: Compute losses $\mathcal{L}_{\text{sup}}, \mathcal{L}_{\text{pl}}, \mathcal{L}_{\text{cons}}, \mathcal{L}_{\text{bd}}$
- 9: Update ϕ with $\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_{\text{pl}}\mathcal{L}_{\text{pl}} + \lambda_{\text{cons}}\mathcal{L}_{\text{cons}} + \lambda_{\text{bd}}\mathcal{L}_{\text{bd}}$
- 10: Update $\theta \leftarrow \alpha\theta + (1 - \alpha)\phi$
- 11: **end while**

III. METHOD

A. Problem Formulation and Overview

Let $\mathcal{D}_l = \{(x_i^l, y_i^l, d_i^l)\}_{i=1}^{N_l}$ denote labeled RGB images, binary masks, and pre-computed monocular depth maps, and let $\mathcal{D}_u = \{(x_i^u, d_i^u)\}_{i=1}^{N_u}$ denote unlabeled RGB-depth pairs. The goal is to learn a segmentation model from $\mathcal{D}_l$ and $\mathcal{D}_u$ under low label ratios. GeoRisk-SPC follows an EMA teacher-student framework [8]. The teacher T_θ generates probability maps p_t from weakly augmented unlabeled images, while the student S_ϕ predicts segmentation maps from strongly augmented images. The teacher parameters are updated as $\theta \leftarrow \alpha\theta + (1 - \alpha)\phi$. The central design is to avoid applying the same pseudo-label rule to every unlabeled pixel. GeoRisk-SPC estimates a risk map R from depth structure and teacher predictions, partitions unlabeled pixels into high-risk and low-risk subsets, and then applies different supervision to the two subsets. The framework contains four components:

- 1) Geometry-aware risk localization identifies unreliable pseudo-label regions using depth discontinuity, teacher uncertainty, and geometry-semantic conflict.
- 2) Region-aware supervision applies hard pseudo-label learning to low-risk pixels and soft consistency learning to high-risk pixels.
- 3) Structural perturbation consistency injects feature perturbations in high-risk regions to learn robustness where pseudo-label errors concentrate.
- 4) The depth-guided geometric encoder injects multi-scale depth and depth-gradient context into RGB features through gated residual modulation.

The student uses a shared encoder and two decoders. The clean decoder produces standard logits z_{clean} , and the perturbed decoder produces logits z_{pert} from risk-perturbed bottleneck features. We evaluate two variants: GeoRisk-SPC-UNet, which uses an RGB-D UNet backbone, and GeoRisk-SPC-DG, which replaces the encoder with a depth-guided geometric encoder. Algorithm 1 summarizes the training loop.

B. Geometry-aware Risk Localization

The risk localization module estimates where teacher pseudo-labels should not be trusted as hard targets. For an unlabeled image x_u with depth map d_u , we first compute a locally normalized depth map within a sliding window of size $w \times w$:

$$d_{\text{rel}} = \frac{d_u - \mu_{\text{local}}(d_u)}{\sigma_{\text{local}}(d_u) + \epsilon} \quad (1)$$

where μ_{local} and σ_{local} are the local mean and standard deviation. Local normalization removes global scale bias and emphasizes relative depth changes within each surgical frame. Depth discontinuity is then computed by the Sobel gradient magnitude:

$$G_d = \|S * d_{\text{rel}}\| \quad (2)$$

where S denotes the Sobel operator. Given teacher probabilities p_t , teacher uncertainty is measured by normalized entropy:

$$U_t = -\frac{1}{\log C} \sum_{c=1}^C p_t^c \log(p_t^c + \epsilon) \quad (3)$$

where C is the number of classes. We also compute a prediction boundary map from the class probability gradients:

$$B_p = \frac{1}{C} \sum_{c=1}^C \|S * p_t^c\| \quad (4)$$

Depth and prediction boundaries are independently min-max normalized to $\bar{G}_d$ and $\bar{B}_p$. Their mismatch defines geometry-semantic conflict:

$$C_{\text{conf}} = |\bar{G}_d - \bar{B}_p| \quad (5)$$

The final risk map combines uncertainty at geometric discontinuities with explicit boundary conflict:

$$R = \bar{U}_t \cdot \bar{G}_d + \lambda_c C_{\text{conf}} \quad (6)$$

where $\bar{U}_t$ is min-max normalized teacher uncertainty and λ_c controls the contribution of geometry-semantic conflict. High-risk and low-risk masks are defined as:

$$M_r = \mathbb{I}[R > \tau_r] \quad (7)$$

$$M_l = \mathbb{I}[\text{conf} > \tau_c] \cdot (1 - M_r) \quad (8)$$

where $\text{conf} = \max_c p_c$ is the teacher prediction confidence.

C. Region-aware Supervision

The supervised loss on labeled images combines Dice and cross-entropy losses:

$$\mathcal{L}_{\text{sup}} = \frac{1}{2}(\mathcal{L}_{\text{dice}} + \mathcal{L}_{\text{ce}}) \quad (9)$$

For unlabeled images, low-risk pixels are high-confidence regions outside the high-risk mask. They receive hard pseudo-label supervision:

$$\mathcal{L}_{\text{pl}} = \frac{1}{|M_l|} \sum_{(i,j) \in M_l} \text{CE}(p_{\text{clean}}^{(i,j)}, \hat{y}^{(i,j)}) \quad (10)$$

where $\hat{y} = \arg \max_c p_i^c$. High-risk pixels do not receive hard pseudo-labels. Instead, the perturbed branch is constrained to remain consistent with the clean branch:

$$\mathcal{L}_{cons} = \frac{1}{|M_r|} \sum_{(i,j) \in M_r} \text{KL}(p_{pert}^{(i,j)} \| \text{sg}[p_{clean}^{(i,j)}]) \quad (11)$$

where $\text{sg}[\cdot]$ denotes stop-gradient. This design prevents high-risk pixels from reinforcing noisy teacher pseudo-labels while preserving a soft learning signal. A boundary consistency term further aligns the spatial gradients of the clean and perturbed predictions in high-risk regions:

$$\mathcal{L}_{bd} = \frac{1}{|M_r|} \sum_{(i,j) \in M_r} |||S * p_{clean}|| - ||S * p_{pert}||| \quad (12)$$

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{sup} + w(t) (\lambda_{pl} \mathcal{L}_{pl} + \lambda_{cons} \mathcal{L}_{cons} + \lambda_{bd} \mathcal{L}_{bd}) \quad (13)$$

where $w(t) = \exp(-5(1-t/T_r)^2)$ is the consistency ramp-up schedule.

D. Structural Perturbation Consistency

Consistency training is most useful when perturbations expose the model to difficult variations. GeoRisk-SPC therefore perturbs bottleneck features only around high-risk regions. The risk mask is downsampled to the bottleneck resolution and used to gate channel dropout and Gaussian noise:

$$f_{pert} = f_{clean} \odot (1 - \hat{M}_r \odot (1 - D_{channel})) + \hat{M}_r \odot \mathcal{N}(0, \sigma^2) \quad (14)$$

where $\hat{M}_r$ is the resized risk mask, $D_{channel} \in \{0, 1\}^{B \times C \times 1 \times 1}$ is a channel-wise binary dropout mask, and σ is the noise standard deviation. The clean branch remains unperturbed and serves as the stop-gradient consistency target. Compared with random feature perturbation, this design concentrates robustness learning on the ambiguous regions identified by geometric risk.

E. Depth-guided Geometric Encoder

The full GeoRisk-SPC-DG model uses DepthGuiderV4 modules at each UNet encoder level. Unlike simple RGB-D concatenation, the encoder processes RGB features and injects depth information through a residual modulation branch. At level l , the module first resizes the depth map to the RGB feature resolution and encodes both depth appearance and depth-gradient geometry:

$$g^l = E_g^l([\nabla_x d^l, \nabla_y d^l, \|\nabla d^l\|]) \quad (15)$$

$$h^l = E_d^l(d^l) \quad (16)$$

where E_g^l and E_d^l are convolutional encoders for geometry and depth. The two streams are mixed to form a depth feature m^l . Local, pooled, and global context branches are then built from m^l :

$$c^l = \omega_1^l c_{local}^l + \omega_2^l c_{pooled}^l + \omega_3^l c_{global}^l \quad (17)$$

where the weights $\omega^l = \text{softmax}(R_s^l([q^l, m^l, g^l]))$ are predicted by a scale router and q^l is a projected RGB feature. The final depth-conditioned residual is:

$$\Delta f^l = \sigma(G^l([F^l([q^l, c^l, q^l \odot \sigma(c^l), |q^l - c^l|]), c^l])) \odot P^l(F^l) \quad (18)$$

and the encoder output is $f_{out}^l = f_{rgb}^l + \Delta f^l$. This formulation lets the network inject depth strongly at boundary-sensitive shallow layers and suppress it where semantic RGB features are sufficient. The same depth maps are also used by the risk localization module, so geometry contributes both to feature extraction and to unlabeled-pixel trust estimation.

IV. EXPERIMENTS

A. Dataset

We evaluate on the EndoVis 2017 [2] surgical instrument segmentation dataset (Task 1), which contains 1800 images from 10 video sequences (1350 training, 450 testing). Task 1 is a binary segmentation task (instrument vs. background). We use 4-fold cross-validation with sequence-based splitting, following the standard protocol. We evaluate under different labeled data ratios: 5%, 10%, 20%, and 40%.

Pre-computed depth maps are generated using a monocular depth estimation model [19]. For GeoRisk-SPC-UNet, depth is concatenated with RGB as an additional input channel. For GeoRisk-SPC-DG, depth is injected through the depth-guided geometric encoder. In both variants, depth discontinuities are used to compute the risk map for unlabeled-pixel supervision.

B. Implementation Details

We implement our method in PyTorch and train on a single NVIDIA RTX 5060 Ti GPU. The key hyperparameters are:

- Backbone: UNet [30] with 16 initial filters
- Optimizer: Adam with learning rate $1e-4$ and weight decay $1e-4$
- Batch size: 24 (12 labeled + 12 unlabeled samples)
- Training iterations: 30,000 with early stopping (patience=9,000 iterations)
- EMA decay rate: $\alpha = 0.99$ for teacher model update
- Depth maps: pre-computed monocular depth estimation (1 channel)
- Risk map parameters: $\tau_r = 0.5$ (risk threshold), $\tau_c = 0.9$ (confidence threshold), $\lambda_c = 0.5$ (conflict weight)
- Perturbation: channel dropout rate=0.3, Gaussian noise std=0.1
- Loss weights: $\lambda_{pl} = 1.0$, $\lambda_{cons} = 1.0$, $\lambda_{bd} = 0.5$
- Data augmentation: weak augmentation (random flip/rotation for teacher) + strong augmentation (RandAugment for student)

C. Main Results

Table I compares GeoRisk-SPC with baselines across different labeled data ratios on EndoVis 2017 (4-fold cross-validation). Methods are sorted by publication year. For UniMatch and SegMatch, we report both our reimplementations and the official code results. GeoRisk-SPC-DG achieves the

best mean Dice among all methods at 5%, 10%, 20%, and 40% labels. Compared to UniMatch (reimpl.), GeoRisk-SPC-DG improves by +2.59% at 5%, +1.46% at 10%, +1.74% at 20%, and +2.34% at 40%. Compared to SegMatch (reimpl.), the improvements are +1.41%, +1.69%, +1.64%, and +2.58% respectively. Against the official implementations, GeoRisk-SPC-DG surpasses UniMatch (official) by +3.88%, +4.09%, +3.46%, and +3.34%, and SegMatch (official) by +3.22% at 5% labels. Notably, GeoRisk-SPC-DG with only 10% labels (88.09%) surpasses SegMatch (reimpl.) with 40% labels (87.20%), demonstrating superior label efficiency through geometry-aware risk supervision.

The base model GeoRisk-SPC-UNet, which uses standard UNet with depth concatenated as input, also outperforms all classical baselines (MT, UAMT, URPC) and surpasses UniMatch at 20% and 40% labels, validating that the risk-conditioned supervision framework is effective even without the depth-guided encoder.

D. Ablation Studies

We conduct component and variant analysis on the EndoVis 2017 dataset with 20% labeled data. Results are shown in Table II. The table evaluates the effect of risk-guided perturbation, boundary consistency, risk-source choices, and depth-guided geometry encoding under the same training protocol.

The architecture comparisons show that depth-guided geometric encoding improves the Mean Teacher baseline from 87.43% to 88.83%, and GeoRisk-SPC-UNet further improves over the RGB-D Mean Teacher baseline. Random masks and removal of boundary consistency reduce performance relative to the full model, supporting the use of spatially targeted perturbation and boundary-aware regularization. DG encoder without risk supervision achieves 89.18%, confirming that the depth-guided architecture itself contributes meaningfully. Threshold sensitivity analysis shows that $\tau_r = 0.3$ yields the best single-threshold result (89.30%), while $\tau_r = 0.7$ degrades performance (88.75%), indicating that a lower risk threshold captures more high-risk regions for perturbation-based consistency. The confidence threshold τ_c is more robust, with both 0.7 and 0.95 performing comparably to the default 0.9. Combining $\tau_r = 0.3$ with $\tau_c = 0.7$ achieves 89.31%, the best overall ablation result, suggesting that relaxing both thresholds simultaneously allows the risk localization mechanism to cover more uncertain regions while maintaining reliable low-risk pseudo-labels.

E. Analysis

1) *Risk Map Visualization*: Figure 2 visualizes the risk maps generated by our method. The high-risk regions (red) correspond to instrument boundaries and ambiguous areas where depth discontinuities coincide with prediction uncertainty. The low-risk regions (green) are primarily in the interior of instruments and background areas where pseudo-labels are reliable. The risk map effectively captures the spatial heterogeneity of pseudo-label quality, enabling targeted supervision strategies.

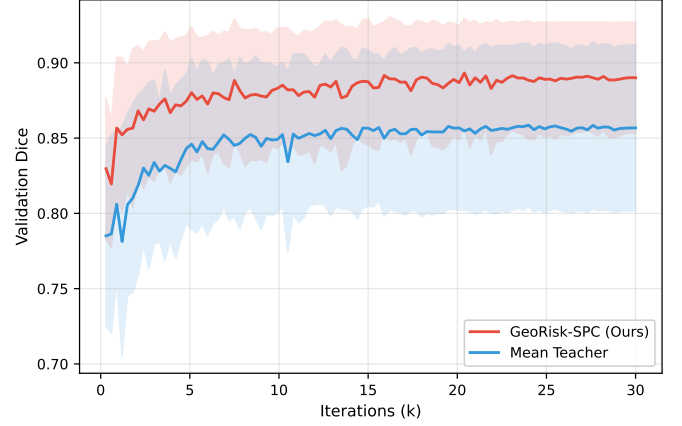


Fig. 1. Training curves of GeoRisk-SPC and Mean Teacher on EndoVis 2017 (40% labeled data, 4-fold average). GeoRisk-SPC converges faster and achieves higher final Dice score. Shaded regions indicate standard deviation across folds.

2) *Effect of Labeled Data Ratio*: Table I shows that GeoRisk-SPC consistently outperforms classical baselines (MT, UAMT, URPC) across different labeled data ratios. Compared to MT, GeoRisk-SPC-UNet achieves +0.77% at 5%, +1.30% at 10%, +1.19% at 20%, and +2.25% at 40%. The advantage grows with more labeled data, as the risk localization mechanism benefits from better teacher predictions. GeoRisk-SPC-DG further extends these gains, surpassing UniMatch and SegMatch at all label ratios.

3) *Cross-fold Consistency*: The standard deviations in Table I show that GeoRisk-SPC-DG achieves competitive variance across folds. At 40% labeled ratio, GeoRisk-SPC-DG has std of 3.45% while MT has 4.94%, suggesting more stable performance. The risk localization mechanism provides more consistent pseudo-label quality across folds, reducing performance variance.

4) *Effect of Depth-guided Encoder*: The depth-guided encoder (GeoRisk-SPC-DG) consistently outperforms the base UNet model (GeoRisk-SPC-UNet) across all label ratios. At 20% labeled ratio, GeoRisk-SPC-DG achieves 88.95% Dice compared to GeoRisk-SPC-UNet's 88.62% (+0.33%). The gap is largest at 10% labels: 88.09% vs 86.88% (+1.21%). At 5% labels, GeoRisk-SPC-DG achieves 86.89% vs 85.43% (+1.46%). The multi-scale gated depth modulation provides richer geometric features that enhance both risk localization accuracy and boundary delineation, with the benefit being more pronounced when labeled data is scarce.

5) *Convergence Analysis*: Figure 1 shows the training curves of GeoRisk-SPC compared to standard Mean Teacher. GeoRisk-SPC converges faster and achieves higher final performance. The region-aware supervision strategy provides more stable training signals, reducing the impact of noisy pseudo-labels during the learning process.

6) *Selective Depth Feature Usage*: Figure 3 visualizes the depth feature activations at each encoder level of the DepthGuiderV4 module across three test samples. Each row corresponds to a different input case, showing the depth encoder output (*depth_feat*) and geometry encoder output

TABLE I
COMPARISON WITH BASELINES ACROSS DIFFERENT LABELED DATA RATIOS (VALIDATION BEST DICE, ALL FOLDS)

Method	Year	5%	10%	20%	40%
Sup-only	–	84.87	85.41	87.51	87.09
MT [8]	2017	84.66 $\pm$ 4.42	85.58 $\pm$ 4.28	87.43 $\pm$ 4.07	86.85 $\pm$ 4.94
UAMT [13]	2019	85.31 $\pm$ 4.10	85.62 $\pm$ 4.04	86.94 $\pm$ 4.39	87.36 $\pm$ 3.50
URPC [14]	2021	85.56 $\pm$ 4.02	87.15 $\pm$ 4.10	87.87 $\pm$ 4.31	87.57 $\pm$ 3.73
CPS [10]	2021	84.50 $\pm$ 4.99	85.90 $\pm$ 4.09	87.36 $\pm$ 4.13	84.50 $\pm$ 3.83
U2PL [11]	2022	85.49 $\pm$ 4.26	86.39 $\pm$ 4.11	87.42 $\pm$ 3.97	88.64 $\pm$ 3.16
UniMatch [12]	2023	84.30 $\pm$ 5.15	86.63 $\pm$ 4.02	87.21 $\pm$ 3.82	87.44 $\pm$ 4.09
UniMatch (official)	2023	83.01 $\pm$ 7.41	84.00 $\pm$ 2.59	85.49 $\pm$ 4.21	86.44 $\pm$ 4.10
MMS [6]	2023	84.07 $\pm$ 4.90	84.92 $\pm$ 4.82	84.33 $\pm$ 5.63	85.63 $\pm$ 6.06
CorrMatch [25]	2024	84.00 $\pm$ 5.11	85.40 $\pm$ 4.26	86.50 $\pm$ 4.18	87.36 $\pm$ 4.02
CW-BASS [31]	2025	84.24 $\pm$ 4.90	85.62 $\pm$ 3.89	87.37 $\pm$ 4.40	–
SegMatch [7]	2025	85.48 $\pm$ 4.72	86.40 $\pm$ 4.46	87.31 $\pm$ 4.51	87.20 $\pm$ 4.31
SegMatch (official)	2025	83.67 $\pm$ 7.10	85.64 $\pm$ 4.69	–	–
GeoRisk-SPC-UNet (ours)	2026	85.43 $\pm$ 5.45	86.88 $\pm$ 4.88	88.62 $\pm$ 4.65	89.10 $\pm$ 4.45
GeoRisk-SPC-DG (ours)	2026	86.89 $\pm$ 4.94	88.09 $\pm$ 4.33	88.95 $\pm$ 4.05	89.78 $\pm$ 3.45

Validation best dice during training. Sup-only uses only labeled data without unlabeled supervision. “(official)” denotes results from the authors’ released code; unmarked rows use our reimplement. Missing entries indicate unavailable completed runs under the same protocol.

TABLE II
COMPONENT AND VARIANT ANALYSIS OF GEORISK-SPC (20% LABELED, ALL FOLDS)

Configuration	Dice	Δ
GeoRisk-SPC-DG (full)	88.95 $\pm$ 4.05	–
w/o $\mathcal{L}_{cons}$	89.22 $\pm$ 4.31	+0.27
w/o $\mathcal{L}_{bd}$	88.90 $\pm$ 4.49	-0.05
Random mask (no risk guidance)	88.76 $\pm$ 4.53	-0.19
w/o risk localization	89.25 $\pm$ 4.31	+0.30
<i>Risk source decomposition</i>		
Depth only (G_d)	89.12 $\pm$ 4.33	+0.17
Uncertainty only (U_t)	88.95 $\pm$ 4.68	+0.00
Conflict only (C_{conf})	89.14 $\pm$ 4.30	+0.19
Depth + Uncertainty	89.03 $\pm$ 4.44	+0.08
<i>Architecture ablation</i>		
MT + RGB-D concatenation	87.43 $\pm$ 4.07	-1.52
MT + DG encoder	88.83 $\pm$ 4.06	-0.12
GeoRisk-SPC w/o DG (UNet)	88.62 $\pm$ 4.65	-0.33
DG encoder w/o risk supervision	89.18 $\pm$ 4.43	+0.23
<i>Threshold sensitivity</i>		
$\tau_r = 0.3$	89.30 $\pm$ 4.31	+0.35
$\tau_r = 0.7$	88.75 $\pm$ 4.66	-0.19
$\tau_c = 0.7$	89.17 $\pm$ 4.53	+0.23
$\tau_c = 0.95$	88.97 $\pm$ 4.46	+0.02
$\tau_r=0.3, \tau_c=0.7$	89.31 $\pm$ 4.13	+0.36

(*geom_feat*) at layers L0–L4.

A striking pattern emerges: depth features are actively utilized at shallow layers (L0–L2) but suppressed to near-zero at deep layers (L3–L4). At L0 (16 channels, 224×224), depth features exhibit strong spatial activations with values up to 2.88, indicating rich geometric information is being extracted. L1 and L2 maintain meaningful activations (max values 1.52 and 1.36 respectively). In contrast, L3 and L4 show near-zero activations (max $< 10^{-4}$), effectively learning to ignore depth input.

This selective usage pattern has a natural interpretation. Shallow layers are responsible for capturing fine-grained details such as edges, textures, and boundaries, precisely where geometric information from depth discontinuities is

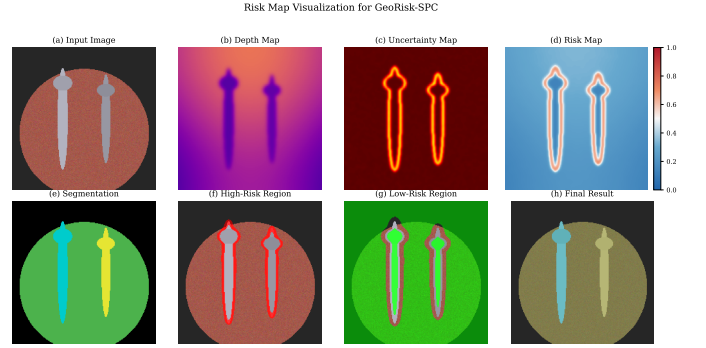


Fig. 2. Visualization of risk maps. From left to right: input image, depth map, risk map, and segmentation result. High-risk regions (red) correspond to instrument boundaries and ambiguous areas.

most valuable for distinguishing instrument edges from tissue boundaries. Deep layers encode high-level semantic features where the category information suffices and depth cues provide diminishing marginal returns. The model learns this “use depth at shallow layers, ignore it at deep layers” strategy autonomously, without explicit supervision to do so.

This finding also validates the design of our risk localization module, which computes depth discontinuity at the input resolution, where the raw geometric signal is strongest, rather than at deep feature levels. The DepthGuiderV4 module further enhances this by fusing depth features at each encoder level, allowing the network to selectively attend to geometric cues where they are most informative.

V. DISCUSSION

A. Why Geometry-conditioned Risk Supervision Works

The effectiveness of our approach stems from two complementary insights. First, pseudo-label reliability is spatially heterogeneous in surgical scenes: instrument boundaries exhibit complex 3D geometry, motion blur, tissue deformation, and specular reflections that make pseudo-labels unreliable.

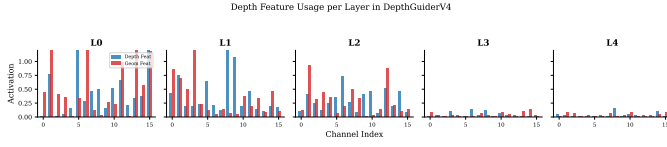


Fig. 3. Depth feature activations per encoder layer in DepthGuiderV4. Each row shows a test sample with the input image overlaid with depth (column 1) and depth encoder outputs at layers L0–L4 (columns 2–6). Shallow layers (L0–L2) show active depth feature extraction, while deep layers (L3–L4) learn to suppress depth contributions (gray regions marked “OFF”). This selective usage pattern indicates that geometric cues are most beneficial for capturing fine-grained boundary details rather than high-level semantics.

Second, this heterogeneity correlates with geometric structure: depth discontinuities from monocular depth estimation coincide with instrument boundaries, providing an external structural cue for identifying unreliable regions.

Once risk is localized, differentiated supervision becomes possible: hard pseudo-labels in reliable regions provide strong learning signals, while soft structural consistency in high-risk regions prevents error propagation. This is more effective than either uniformly trusting all pseudo-labels (which propagates errors) or uniformly downweighting them (which wastes reliable signals).

B. Computational Overhead

GeoRisk-SPC introduces minimal computational overhead. The risk localization module adds approximately 5% to the training time compared to standard Mean Teacher. The dual-decoder architecture increases memory usage by about 30%, but this is offset by the improved convergence speed (fewer iterations needed to reach optimal performance).

C. Limitations

Our method has several limitations:

- **Depth quality dependency:** The effectiveness of risk localization depends on the quality of pre-computed depth maps. Poor depth estimation may lead to inaccurate risk maps. Future work could explore using multiple depth estimators or uncertainty-aware depth fusion.
- **Threshold sensitivity:** The performance is sensitive to the risk threshold τ_r (89.30% at 0.3 vs. 88.75% at 0.7) while the confidence threshold τ_c is more robust (Table II). Adaptive thresholding strategies could further improve robustness.
- **Dataset scope:** We evaluated on EndoVis 2017 (binary segmentation). Validation on multi-class surgical datasets and other surgical scenarios is needed to confirm generalizability.
- **Computational overhead:** The dual-decoder architecture increases memory usage by approximately 30% compared to standard Mean Teacher, though this is offset by improved convergence speed.

VI. CONCLUSION

We proposed GeoRisk-SPC, a geometry-conditioned pseudo-label risk supervision framework for semi-supervised

surgical instrument segmentation. Our key insight is that pseudo-label reliability is spatially heterogeneous and correlates with geometric structure: depth discontinuities from monocular depth estimation reveal where pseudo-labels are unreliable. Based on this insight, we design a risk localization mechanism that partitions unlabeled pixels into low-risk and high-risk regions, applying hard pseudo-labels to reliable regions and soft structural consistency to ambiguous regions.

The full model GeoRisk-SPC-DG incorporates a depth-guided geometric encoder with multi-scale gated residual modulation for enhanced geometric feature extraction. Experiments on EndoVis 2017 demonstrate that GeoRisk-SPC-DG achieves 89.78% Dice at 40% labels, 88.09% at 10% labels, and 86.89% at 5% labels, outperforming strong baselines UniMatch and SegMatch across all settings. Notably, GeoRisk-SPC-DG with 10% labels surpasses SegMatch with 40% labels, demonstrating that geometry-aware risk supervision enables superior label efficiency.

Future work includes extending our method to 3D surgical scene segmentation, exploring adaptive thresholding strategies for risk map generation, and validating on additional surgical datasets to confirm generalizability.

REFERENCES

- [1] L. Maier-Hein, S. S. Vedula, S. Speidel, N. Navab, R. Kikinis, A. Park, M. Eisenmann, H. Feussner *et al.*, “Surgical data science for next-generation interventions,” *Nature Biomedical Engineering*, vol. 1, no. 9, pp. 691–696, 2017.
- [2] M. Allan *et al.*, “2017 robotic instrument segmentation challenge,” in *arXiv preprint arXiv:1902.06426*, 2019.
- [3] T. Rueckert, D. Rueckert, and C. Palm, “Methods and datasets for segmentation of minimally invasive surgical instruments in endoscopic images and videos: A review of the state of the art,” *Computers in Biology and Medicine*, vol. 169, p. 107929, 2024.
- [4] H. Peng, S. Lin, D. King, Y.-H. Su, W. M. Abuzaid, R. A. Bly, K. S. Moe, and B. Hannaford, “Reducing annotating load: Active learning with synthetic images in surgical instrument segmentation,” *Medical Image Analysis*, vol. 97, p. 103246, 2024.
- [5] E. Colceoni, R. Sanchez Matilla, I. Luengo, and D. Stoyanov, “Guided image generation for improved surgical image segmentation,” *Medical Image Analysis*, vol. 97, p. 103263, 2024.
- [6] A. Lou, K. Tawfik, X. Yao, Z. Liu, and J. Noble, “Min-max similarity: A contrastive semi-supervised deep learning network for surgical tools segmentation,” *IEEE Transactions on Medical Imaging*, vol. 42, no. 10, pp. 2832–2841, 2023.
- [7] M. Wei, C. Budd, L. C. Garcia-Peraza-Herrera, R. Dorent, M. Shi, and T. Vercauteren, “Segmatch: semi-supervised surgical instrument segmentation,” *Scientific Reports*, vol. 15, no. 1, 2025.
- [8] A. Tarvainen and H. Valpola, “Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results,” in *NeurIPS*, 2017, pp. 1195–1204.
- [9] D. Berthelot *et al.*, “Mixmatch: A holistic approach to semi-supervised learning,” in *NeurIPS*, 2019, pp. 5049–5059.
- [10] X. Chen, Y. Yuan, G. Zeng, and J. Wang, “Semi-supervised semantic segmentation with cross pseudo supervision,” in *CVPR*, 2021, pp. 2613–2622.
- [11] Y. Wang, H. Wang, Y. Shen, J. Fei, W. Li, G. Jin, L. Wu, R. Zhao, and X. Le, “Semi-supervised semantic segmentation using unreliable pseudo-labels,” in *CVPR*, 2022, pp. 4238–4247.
- [12] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, “Revisiting weak-to-strong consistency in semi-supervised semantic segmentation,” in *CVPR*, 2023, pp. 7236–7246.
- [13] L. Yu *et al.*, “Uncertainty-aware self-ensembling model for semi-supervised 3d left atrium segmentation,” in *MICCAI*, 2019, pp. 605–613.
- [14] X. Chen *et al.*, “Semi-supervised medical image segmentation with uncertainty rectified pyramid consistency,” in *MICCAI*, 2021, pp. 53–63.

- [15] J. Liu, C. Desrosiers, D. Yu, and Y. Zhou, "Semi-supervised medical image segmentation using cross-style consistency with shape-aware and local context constraints," *IEEE Transactions on Medical Imaging*, vol. 43, no. 4, pp. 1449–1461, 2024.
- [16] C. Xu, T. Zhang, D. Zhang, D. Zhang, and J. Han, "Deep generative adversarial reinforcement learning for semi-supervised segmentation of low-contrast and small objects in medical images," *IEEE Transactions on Medical Imaging*, vol. 43, no. 9, pp. 3072–3084, 2024.
- [17] Y. Yuan, X. Wang, X. Yang, and P.-A. Heng, "Effective semi-supervised medical image segmentation with probabilistic representations and prototype learning," *IEEE Transactions on Medical Imaging*, vol. 44, no. 3, pp. 1181–1193, 2025.
- [18] Q. Zeng, Y. Xie, Z. Lu, M. Lu, Y. Wu, and Y. Xia, "Segment together: A versatile paradigm for semi-supervised medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 44, no. 7, pp. 2948–2959, 2025.
- [19] R. Ranftl, A. Bochkovskiy, and V. Koltun, "Vision transformers for dense prediction," in *ICCV*, 2021, pp. 12 159–12 168.
- [20] L. Mi, Z. Zhang, J. Zhang *et al.*, "Depth adaptive deep neural network for semantic segmentation," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 11, pp. 2590–2602, 2017.
- [21] Y. Zhu *et al.*, "Depth-aware attention network for semantic segmentation," *Pattern Recognition*, vol. 100, p. 107146, 2020.
- [22] B.-W. Yin, J.-L. Cao, D. Xu, M.-M. Cheng, and Q. Hou, "Dformer++: Improving rgbd representation learning for semantic segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–14, 2026.
- [23] E. Colleoni and D. Stoyanov, "Robotic instrument segmentation with image-to-image translation," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 935–942, 2021.
- [24] L. Yang, Z. Zhao, and H. Zhao, "Unimatch v2: Pushing the limit of semi-supervised semantic segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 4, pp. 3031–3048, 2025.
- [25] B. Sun, Y. Yang, L. Zhang, M.-M. Cheng, and Q. Hou, "Corrmatch: Label propagation via correlation matching for semi-supervised semantic segmentation," in *CVPR*, 2024, pp. 3097–3107.
- [26] A. Das, C. Gautam, H. Cholakkal, P. Agrawal, F. Yang, R. Savitha, and Y. Liu, "Decoupled training for semi-supervised medical image segmentation with worst-case-aware learning," in *MICCAI*, 2024, pp. 45–55.
- [27] Q. Liu *et al.*, "Surgivisor: Transformer-based semi-supervised surgical instrument segmentation with uncertainty-guided supervision," *Computerized Medical Imaging and Graphics*, vol. 112, p. 102345, 2024.
- [28] D.-H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," *ICML Workshop*, 2013.
- [29] B. Zhang *et al.*, "Flexmatch: Boosting semi-supervised learning with curriculum pseudo labeling," in *NeurIPS*, 2021, pp. 18 769–18 781.
- [30] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *MICCAI*, 2015, pp. 234–241.
- [31] E. Tarubinga, J. Kalafatovich, and S.-W. Lee, "Cw-bass: Confidence-weighted boundary-aware learning for semi-supervised semantic segmentation," in *IJCNN*, 2025, pp. 1–8.